

AI-Fiscal v2.0 — proposed methodology

Draft for review

John Iselin and Ryan Nunn, The Budget Lab at Yale

October 2, 2026

Contents

1. The question	2
2. Overview: from the AI shock to tax outputs	2
3. Modules	4
3.0 Inputs and baseline [D6, D13]	4
3.1 Production accounts [D1, D2, D3]	5
3.2 Labor distribution (existing S0/S2/S3) [D8]	6
3.3 Capital by legal form and the entity tax [D4]	6
3.4 Payout [D7]	6
3.5 Ownership and wrappers [D9] — <i>new</i>	7
3.6 Household income and asset state [D10, D11] — <i>partly new</i>	7
3.7 Reconciliation to the tax file [D5]	8
3.8 Microsimulation	8
3.9 Outputs [D12, D13]	8
4. Scenarios and sensitivities	8
5. Validation	9
6. Building on existing Budget Lab models	9
6.1 Reuse directly	10
6.2 Adapt	11
6.3 Build new	11
6.4 Effect on the plan	12
7. Out of scope for v2.0	12
8. Questions for review	12
Appendix A — the \$100 design test	12

Status. Draft for review, 2026-10-02. Nothing in this document is a SET decision. Every choice cites its entry in [v2_decisions.md](#) ([D1]–[D13]), and “reference case” means the default proposed there — the one to argue against. The design follows the September 23 review ([v2_review_and_roadmap.md](#)). Where this document and [v2_flow_network.md](#) disagree, this one reflects the review and the network has not yet been revised.

What to review. (1) whether the estimand in §1 is the question we want to answer; (2) the reference-case choices in §3, module by module; (3) the reuse plan in §6 — what we take from existing Budget Lab models versus build; (4) the questions collected in §8.

1. The question

Given an annual path for aggregate output and its split between labor and capital, what happens to federal individual income, payroll, and corporate income taxes, and to the distribution of household income, under current law and explicitly stated transmission assumptions?

v1.0 answered a narrower version: it sized the capital increment on the tax-return base ($X = g_K Y_0^K$, realized income only), so realization, ownership, and entity composition were fixed by construction, and CIT entered as a single macro wedge. v2.0 sizes the shock **upstream** in the national accounts and follows each dollar to a named destination — a federal tax, a household (as cash or as accruing value), a retirement balance, a tax-exempt holder, or a foreign owner.

The research contribution is the **explanation of why the tax system captures different amounts of an otherwise identical aggregate gain**: factor shares, legal form, ownership, tax preferences, and timing, each quantified. v2.0 does *not* forecast AI's aggregate effect (it takes the Karger et al. path as given) and is not a general-equilibrium model.

The scenario is measured **relative to the CBO baseline** [D6], which already embeds an AI assumption (+0.1pp/yr productivity); it is not an AI-versus-no-AI counterfactual. Publication focuses on **2030**, with annual accounting from the start year; no ten-year score in v2.0 [D13].

2. Overview: from the AI shock to tax outputs

The diagram has two halves separated by the dashed rule.

Above the rule, everything is an aggregate annual ledger. The shock enters as a GDP level and a labor-share path (inputs, top). Module 1 turns these into changes in labor income, capital income by channel, and named non-factor items, on one consistent accounting identity. Labor goes straight to the microdata. Capital is split by legal form; C-corporation profits pay CIT (counted once, here — the dashed blue line), then split into dividends and retained income. Module 4 assigns every dollar of distributable capital income to an owner and wrapper. Module 5 turns owner-level flows into three objects: cash income, an asset ledger of unrealized and realized gains, and retirement balances with withdrawals. Whatever is foreign-owned, exempt, Roth, or deferred is reported as a destination and never written onto tax returns.

Below the rule is the microsimulation. Module 6 reconciles the taxable aggregates to the tax file's concepts; module 7 allocates them to tax units with the v1.0 allocators, writes counterfactual records, and runs Tax-Simulator. Outputs are federal receipts by instrument, two distributional panels, and a flow reconciliation that accounts for every dollar of ΔY .

A Mermaid version of the same structure (renders on GitHub):

flowchart TB

```
CBO["CBO baseline · D6"] --> PA
AI["AI path: GDP level + NFB labor share · D1"] --> PA
MA["macro adapter (opt.)"] -.-> AI
PA["1 Production accounts Y = L + K + Q · D1 D2 D3"]
```

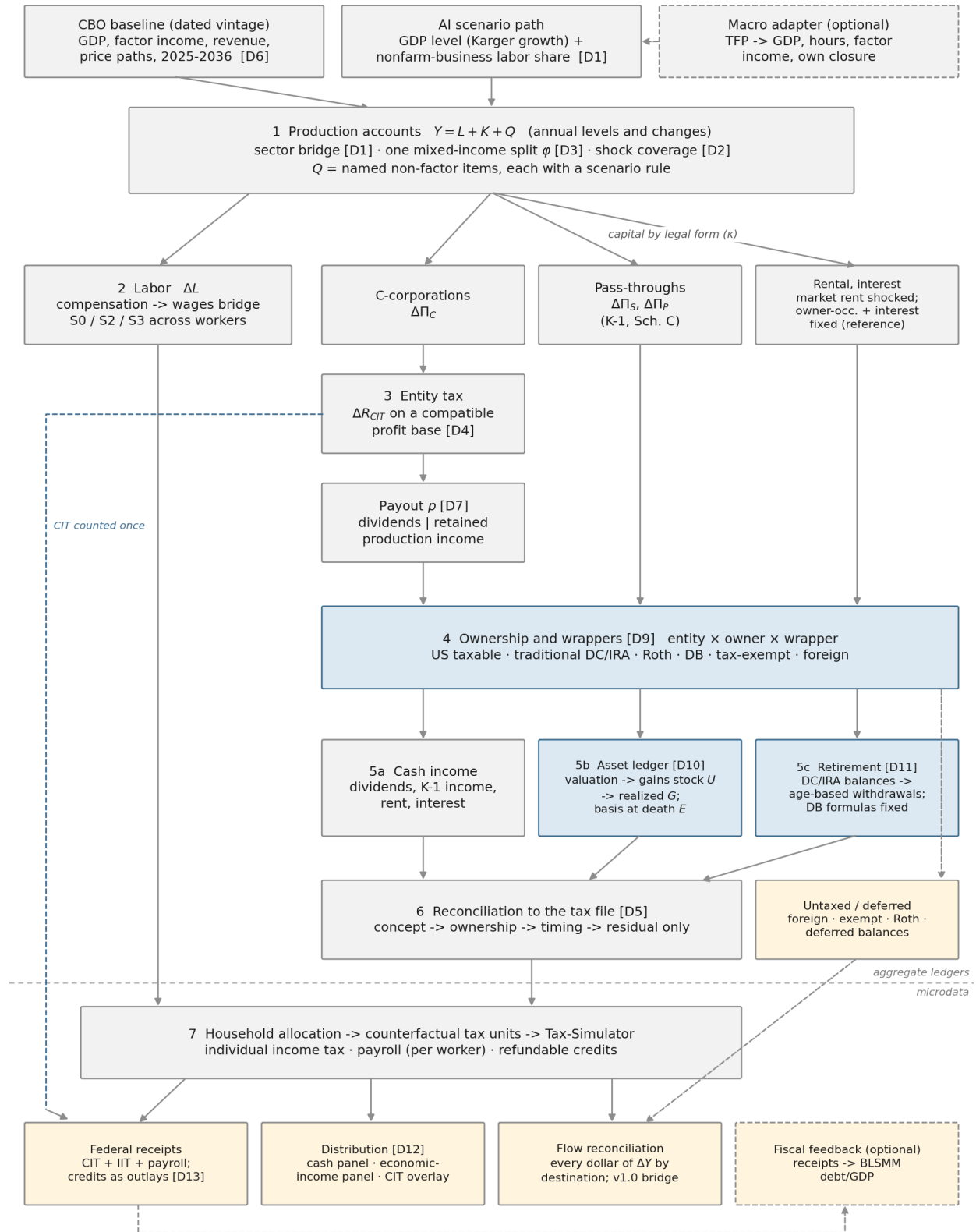


Figure 1: Proposed v2.0 transmission from the AI shock to federal tax and distributional outputs. Blue boxes are stages the September review added; cream boxes are outputs; dashed boxes are optional.

```

PA --> LAB["2 Labor  $\Delta L \rightarrow$  wages  $\rightarrow S0/S2/S3$ "]
PA --> CC["C-corp  $\Delta \Pi_C$ "] --> CIT["3 Entity tax  $\Delta R_{CIT} \cdot D4$ "] --> PAY["payout  $p \cdot D7$ "]
PA --> PT["pass-throughs  $\Delta \Pi_S, \Delta \Pi_P$ "]
PA --> RI["rental, interest"]
PAY & PT & RI --> OWN["4 Ownership  $\times$  wrapper  $\cdot D9$ "]
OWN --> CASH["5a cash income"]
OWN --> AL["5b asset ledger  $\cdot D10$ "]
OWN --> RET["5c retirement  $\cdot D11$ "]
OWN --> UNT["untaxed / deferred destinations"]
CASH & AL & RET --> REC["6 Reconciliation  $\cdot D5$ "]
REC --> MS["7 allocation  $\rightarrow$  tax units  $\rightarrow$  Tax-Simulator"]
LAB --> MS
MS --> REV["federal receipts  $\cdot D13$ "]
CIT -. "counted once" .-> REV
MS --> DIST["distribution panels  $\cdot D12$ "]
MS --> FLOW["flow reconciliation"]
UNT --> FLOW
REV --> BLSMM["BLSMM debt/GDP (opt.)"]

```

3. Modules

Notation: superscripts B (baseline) and S (scenario); $\Delta x = x^S - x^B$; t indexes calendar years. All ledgers are annual and nominal. Each module carries **identical rules in baseline and scenario**, and outputs are differences — no module applies a nonnegativity constraint to a delta.

3.0 Inputs and baseline [D6, D13]

Input	Reference case	Source
Baseline macro path, 2025–2036	CBO February 2026 economic projections: GDP, wages and salaries, domestic economic profits, proprietors' income (and rental, interest, dividends if carried)	CBO pub 62105 + 10-year projections file (data-gathering 2.1–2.3)
Baseline revenue path	CBO revenue by source, FY 2025–2036	data-gathering 2.2
AI GDP path	real GDP level path from Karger Table 19 growth medians (economists' group, pinned vintage [D1])	NBER w35046
AI labor-share path	Karger Table 39 medians, nonfarm business sector, same respondent group	NBER w35046

Input	Reference case	Source
Prices	one common price path for the controlled real-growth experiment; CPI and wage indexing kept as separate inputs	CBO
Policy	current law at the Tax-Simulator vintage	Tax-Simulator

The macro adapter (dashed box) is an optional alternative entry point: a user who wants to supply TFP rather than GDP routes it through a model with its own closure (employment, investment, factor incomes), which then supplies the GDP and factor paths. TFP and GDP multipliers are never applied to the same base.

What changes from v1.0: v1.0 used a single 2030 endpoint with positional growth keys; v2.0 is year-indexed throughout (build plan 1.2).

3.1 Production accounts [D1, D2, D3]

The core identity, for each year and for levels and changes alike:

$$Y_t = L_t + K_t + Q_t$$

with all three terms on the same domestic/national, gross/net convention. Q is not a residual: each component is named (taxes on production and imports less subsidies, business current transfers, government-enterprise surplus, consumption of fixed capital if gross, statistical discrepancy) and carries a scenario rule — held fixed in levels, scaled with output, or shocked.

Sector bridge [D1]. The survey’s labor share is for the nonfarm business sector (NFB). Write $Y = Y^{NFB} + Y^O$. In the reference case the AI-driven GDP change lands in NFB in proportion to its baseline share, the survey share applies within NFB, and the other sectors keep their baseline factor shares:

$$\Delta L_t^{NFB} = \theta_t^S Y_t^{NFB,S} - \theta_t^B Y_t^{NFB,B}, \quad \Delta K_t^{NFB} = \Delta Y_t^{NFB} - \Delta L_t^{NFB} - \Delta Q_t^{NFB}$$

An alternative national mapping is a required sensitivity. Whether θ_t^B is today’s share or CBO’s projected share is part of D1.

Mixed income [D3]. Proprietors’ income is split with one parameter ϕ (capital) and $1-\phi$ (labor), **identically** in the macro accounts and the micro file (`sole_prop`, `farm`, `S-corp` lines). The review’s correction: v2’s draft network added $\phi \cdot \text{PI}$ to labor, which only works at $\phi = 0.5$; restoring the labor half takes the unexplained residual from 13.1% to 8.9% of national income on the 2024 receipt.

Shock coverage [D2]. The accounts stay complete; the shock is assigned to channels. Reference case: business capital income shocked; owner-occupied housing (imputed rent) unshocked; net interest on a fixed financing-cost path. Market rental and a broader interest response are sensitivities.

Labor to wages. ΔL (compensation) becomes taxable wages through a compensation-to-wages bridge (employer social insurance, pension and health contributions), held at baseline ratios in the reference case.

Acceptance: a baseline table and one toy shock that reconcile sector coverage, GDP/NI, compensation/wages, mixed income, and each named Q component; the zero shock gives zero deltas everywhere.

3.2 Labor distribution (existing S0/S2/S3) [D8]

Unchanged in kind from v1.0, now fed an annual ΔL from §3.1: S0 proportional, S2/S3 compressive/expansive log-affine maps. Two changes: (1) the dispersion parameter is decoupled from GDP growth enough to run a pure-inequality case at unchanged GDP; (2) nonpositive labor incomes get explicit treatment (v1.0's $y_{-1} == 0$ inert class). Payroll tax is computed per worker. **Displacement, occupation exposure, and UI are deferred** [D8]; the output contract keeps worker and tax-unit IDs so they can be added.

3.3 Capital by legal form and the entity tax [D4]

Business capital income splits by legal form:

$$\Delta K^{bus} = \Delta \Pi_C + \Delta \Pi_S + \Delta \Pi_P$$

with reference shares from the 2024 receipt (C-corp 0.497 of the broad base); a C-corp/rent-concentrated split is sensitivity 2 (Q2.4).

CIT, two objects [D4].

- *Baseline bridge.* Economic profits $\rightarrow$ federal CIT receipts on **compatible concepts** — a coarse version of CBO's corporate method (pub 59436): profits without IVA/CCAdj, less foreign and loss-firm adjustments, less deductions, times the statutory rate, less credits; liabilities converted to fiscal-year receipts. This fixes the denominator mismatch the review found (14.7% was computed on profits without IVA/CCAdj but applied to a base with them: $\approx \$442B$, not $\$492B$).
- *Incremental response.* $\Delta R_{CIT} = \tau^m \Delta \Pi_C$, with τ^m built from separately labeled assumptions on profit composition (normal return vs rent), investment deductions (permanent 100% bonus depreciation after 2025-01-19), loss utilization, and credits. The reference rule is labeled as such; sensitivity 5 brackets it.

CIT is added **once**, here, and bypasses the microsimulation (where the existing Tax-Simulator CIT baseline is carried unchanged).

Pass-throughs. No entity tax; taxable distributive share flows to owners every year whether or not cash is paid. Labor/capital classification stays separate from legal tax treatment — SE tax and QBI are applied by the calculator. Pass-through earnings raise basis and are never taxed again as a retained-earnings gain.

3.4 Payout [D7]

$$D = p \cdot \Pi_C^{AT}, \quad RE = (1 - p) \cdot \Pi_C^{AT}, \quad \Pi_C^{AT} = \Delta \Pi_C - \Delta R_{CIT} - \text{other entity taxes}$$

p is measured on a matched universe (that universe's dividends over its after-tax profits), not personal dividends over all corporate profits. RE is **retained production income**, not a capital gain; any

effect on equity value goes through the asset ledger (§3.6). The v1.0 0.30/0.70 public-equity split in `asset_to_income_map.csv` is retired in favor of p. Buybacks are deferred but get their own payout/financing category.

3.5 Ownership and wrappers [D9] — *new*

An entity $\times$ owner $\times$ wrapper matrix $\omega_{e,o}$ with $\sum_o \omega_{e,o} = 1$ for every entity type e. Owner destinations in the reference case: US taxable households; traditional DC/IRA; Roth; DB pensions; tax-exempt institutions; foreign owners. Flows to owner o:

$$F_o = \sum_e \omega_{e,o} (D_e + RE_e + \Pi_e^{PT})$$

Intermediaries (funds, insurers) are looked through with one consistent rule. Foreign and exempt flows are **reported destinations**, never renormalized onto PUF households. Calibration: Financial Accounts sector holdings (note the June 2026 table renumbering), SCF, IRA and pension aggregates; aggregate shares with sensitivities, no multinational microsimulation.

Two conservation tests, not one: the sum over all owner destinations equals the distributable flow; the PUF allocation equals the US taxable-household flow.

3.6 Household income and asset state [D10, D11] — *partly new*

5a Cash income. Dividends, taxable distributive shares, market rent, interest received by US taxable owners.

5b Asset ledger [D10]. Two ledgers, linked:

- the *production ledger* (§§3.1–3.5) carries current income;
- the *asset ledger* carries claim values, basis, unrealized gains U, realizations G, and gains removed by basis step-up at death E:

$$U_{t+1} = U_t + A_t - G_t - E_t$$

where A_t is valuation accrual. Reference valuation rule: no announcement revaluation (accrual tracks retained income — labeled as an assumption, not an identity); a capitalization case that values a specified persistent after-tax cash-flow change is the alternative. Realization G_t comes from either a hazard on U_t or a distributed lag on A, calibrated on matched asset universes against multi-year realizations. Order within the year: accrual, then sales, then deaths. Gains stay **outside** the GDP identity. The CRS 61% figure is a long-run realizations-to-accruals ratio, not an annual hazard or a lifetime probability; the 0.84 “lifetime factor” is dropped.

5c Retirement [D11]. Incremental traditional DC/IRA balances accumulate the wrapper’s share of returns and pay out through an age-sensitive withdrawal rule (or a documented aggregate approximation); Roth separate and untaxed on qualified withdrawal; DB benefit formulas fixed in the near term, with sponsor/funding gains reported separately. Balances live in age $\times$ owner $\times$ wrapper cells, not in a pretend household panel. The v1.0 R1 cascade is reused only to allocate the resulting taxable withdrawals across units.

3.7 Reconciliation to the tax file [D5]

Applied in a fixed order, each step with a named purpose:

1. **Concept and coverage** — imputed rent out; NIPA net interest → household interest; dividend concept (confirm whether Tax-Data's `div_ord` includes qualified dividends).
2. **Ownership and wrappers** — already done in §3.5.
3. **Timing and realization** — already done in §3.6.
4. **Measurement residual** — a final factor that covers *only* the gap left after 1–3, by channel, on comparable universes and years. Additive where denominators are near zero or change sign.

The historical SOI/NIPA ratio is evidence for step 4, not a multiplier on top of steps 2–3 (it already contains them).

3.8 Microsimulation

Reuses the v1.0 machinery: across-unit allocation by each channel's holdings base, within-unit mapping to tax lines, the counterfactual writer (06), and Tax-Simulator via `callr` (07). New writer columns as needed (market rental; per-channel dispatch). Tax-Simulator's fiscal-year adjustment drops the earliest year, so each reported year needs its adjacent tax-year file — to be confirmed against the vintage in week 1.

3.9 Outputs [D12, D13]

- **Federal receipts by instrument:** $\Delta R = \Delta R_{CIT} + \Delta R_{IIT} + \Delta R_{payroll}$, with refundable-credit outlays reported separately before any net fiscal measure. Revenue-to-GDP change $\Delta z = (\Delta R/Y_0 - z_0 g)/(1 + g)$ shown alongside dollar levels.
- **Distribution [D12]:** (a) disposable cash income; (b) accrual-based economic income that attributes retained income and deferred returns once. Valuation changes reported separately. CIT is already deducted before allocation, so no second burden deduction; alternative incidence conventions (OTA, TPC, CBO) as a reconciled overlay. Households ranked by baseline income.
- **Flow reconciliation:** every dollar of ΔY by destination (federal tax by instrument, household cash, accruing value, retirement balances, exempt, foreign, Q).
- **v1.0 comparison:** headline change decomposed by mechanism (source corrections, base, entity tax, ownership, realization, retirement, allocation, income concept), in a fixed order with the interaction remainder shown.
- **Optional:** debt/GDP through BLSMM (§6).

4. Scenarios and sensitivities

Reference grid. Keep the three Karger growth cases (slow / moderate / rapid) and the changing-share vs fixed-share comparison, with S0 as the reference labor distribution and S2/S3 as distributional variants.

Structural sensitivities (the review's list, run as local perturbations around the reference set, then two or three coherent combined cases — scenario ranges, not probability intervals):

#	Sensitivity	Decision
1	Survey group, sector-to-national mapping, baseline-share path	D1
2	Economy-wide vs C-corp/rent-concentrated marginal profits	Q2.4
3	Resident vs foreign; traditional / Roth / exempt proportions	D9
4	Gains: announcement timing, realization schedule, matched-coverage calibration	D10
5	CIT: normal-return/investment- heavy vs rent-heavy; loss utilization	D4
6	Household allocation: wealth- vs income-based; top tail; S2/S3	—
7	Retirement withdrawal response; DB treatment; prices/indexing	D11, D6

5. Validation

Four kinds of evidence, kept separate:

1. **Accounting.** Every identity holds in baseline and scenario (production, entity, ownership, household allocation, asset state); zero shock → zero deltas; every dollar excluded from one base appears in a named destination; CIT counted once. These are blocking.
2. **Implementation.** Row-order invariance; scenario runs leave the baseline untouched; unknown channels fail loudly; single-channel tests for qualified/ordinary dividends, SE income, rental losses, loss carryforwards, NIIT, QBI, per-worker payroll caps.
3. **Economic.** Baseline levels against matched-year, matched-concept benchmarks (SOI, BEA, CBO); scenario deltas checked against their mechanism, not against CBO levels. Calibration to CBO is not validation against CBO. Gaps get documented dispositions, not automatic failures.
4. **Publication.** Reproduce v1.0 on its original vintage; decompose the v1.0 → v2.0 change by mechanism. Matching the old headline is not a completion criterion.

The review's \$100 example (Appendix A) and its all-foreign, all-retirement, zero-dividend, and intermediary variants are the module tests for §§3.3–3.5.

6. Building on existing Budget Lab models

Short answer. The *top* of the pipeline (the CBO baseline) and the *bottom* (tax calculation, CIT incidence, debt/GDP) can come largely from existing Budget Lab models. The *middle* — the stages that

are the point of v2 (factor-share shock, legal form, payout, ownership, realization, retirement timing) — has no Budget Lab equivalent and must be built. No existing Budget Lab model has a labor-share or C-corp-vs-pass-through mechanism.

Checked 2026-10-02 against the public Budget-Lab-Yale repos and the local clones of Tax-Simulator (state-tax) and Tax-Data. Items marked *unverified* were not confirmed against source.

6.1 Reuse directly

Macro-Projections → the baseline module (§3.0) [D6]. [Budget-Lab-Yale/Macro-Projections](#) already ingests CBO’s 10-year economic projections, revenue projections, and long-term outlook, and writes projections.csv with, by year: GDP (calendar and fiscal), **compensation, wages, proprietors’ income (farm/nonfarm), rent, interest, personal dividends, corporate profits (corp_profits_adj)**, prices, rates, and **revenue by source (rev_iit, rev_payroll, rev_corp, ...)**. Years past CBO’s window are extended at constant GDP shares. Tax-Simulator and Tax-Data already read the same vintage (v3/2026022522, pinned in Tax-Simulator’s interface_versions.yaml).

- *What it saves:* most of the Phase 0 data-gathering §2 pulls (2.1–2.3), the hand-built cbo_baseline.csv of build step 1.1, and the interpolated 3,200 / 470 benchmark rows (settles build-plan 0.10).
- *Why it’s better than a hand pull:* the AI-Fiscal baseline, Tax-Data’s aging, and Tax-Simulator’s CIT level would all sit on **one vintage by construction**.
- *Check before use:* whether corp_profits_adj is domestic economic profits (with IVA/CCAdj) or a broader concept — this is exactly the denominator D4 needs to get right. *Unverified*.
- *Doesn’t cover:* realizations, the S-corp split, NIPA 1.12 payout / undistributed profits, an NFB labor share.

Tax-Data resources/cbo_1040.csv → reconciliation and gains [D5, D10]. CBO projections of 1040 line items, 2019–2036 (wages, taxable interest + ordinary dividends, qualified dividends, taxable gains, pass-through, pensions, Social Security, UI). This is the PUF-side baseline the reconciliation residual (§3.7 step 4) is measured against, and CBO’s realized-gains path for calibrating §3.6. Its source publication isn’t documented in the repo — *provenance unverified*. Note the route **not** to take: defining an “AI” Macro-Projections vintage and re-aging the PUF through Tax-Data would move lines uniformly with no ownership or distributional allocation, which defeats the purpose.

Tax-Simulator → the calculator and a CIT-incidence rule [D12]. Already the calculator (§3.8). Two further pieces are reusable:

- `src/data/post_processing/distribution.R:232` already allocates corporate-tax changes between labor and capital income, with the labor share phasing from 0 to 20% over ten years (:484). That gives a **Budget Lab house incidence convention** to sit alongside OTA / TPC / CBO in the D12 overlay, and code to adapt rather than write.
- `revenue.R` confirms that `revenues_corp_tax` is a Macro-Projections level pass-through (:52), so v2’s incremental CIT is added outside it, once.
- **Conflict to resolve in D3:** the same module defines labor income as $wages + 0.8 \times (\text{sole_prop} + \text{part_scorp} + \text{farm})$ and capital as the other 0.2 (`distribution.R:140–141`). That is a third mixed-income convention next to the macro receipt’s $\phi = 0.5$ and AI-Fiscal’s Saez–Zucman split. The distributional panels should use the same ϕ as the production accounts, or say why not.

BLSMM → fiscal feedback, downstream only. The [Small Macro Model \(code\)](#) is an annual FY2026–35 model anchored to CBO: output gap, Okun’s law, Phillips curve, Taylor rule, term premium, effective

rate on debt, receipts as a share of GDP, outlays by CBO rules of thumb. It has **no factor-income, profits, or tax-composition detail**, so it cannot size the shock. It is the right consumer of v2’s output, and v1.0 already uses it that way (code/15_blsmm_debt_gdp.R: revenue/GDP delta → BLSMM receipts, plus a productivity bump calibrated to Karger growth). Its companion article on AI says receipts/GDP is “minimally affected” under rules of thumb and flags that AI could interact with the tax system “because capital is more lightly taxed than labor” — the question AI-Fiscal answers.

- *Proposed change for v2:* BLSMM now ships its own Karger-based AI scenarios (scenarios/inputs/ai_slow.R, ai_rapid.R, ai_s1_productivity.R, ai_s2_prod_lf.R, ...). Have v2 consume those productivity paths rather than re-solving for its own bump in 15, so BLSMM’s published AI article and AI-Fiscal describe the same scenario. Whether BLSMM’s GDP path should also *be* v2’s AI GDP input (the “macro adapter” in §3.0) is question 6 below.

6.2 Adapt

Model	What it could supply	v2 stage	Effort / caveat
Cost-Recovery-Simulator (feeds Tax-Simulator)	Depreciation / expensing revenue deltas, C-corp vs pass-through	Incremental CIT τ^m , investment-deduction component [D4]	Needs an AI-investment input path; replaces a hand receipt for bonus depreciation
CBO rules-of-thumb code in Tariff-Model (src/07_calculate_dynamic_revenue.R)	Growth → revenue feedback	Optional macro feedback	Only if v2 wants feedback outside BLSMM
FRB/US helpers (Budget-Lab-Yale/FRBUS)	Dynamic personal and corporate tax bases	Macro adapter (later release)	Whether the setup can take a factor-share shock is <i>unverified</i>
AI_Exposure_Metrics (private), AI-Employment-Model (public)	SOC-level exposure	Labor extension (deferred, D8)	Inferred from names and file lists only; confirm with the tracker team
Buy-Borrow-Die-Estimates, Estate-Tax-Distribution	Unrealized gains by holder; inheritances	Basis at death E, gains stock U [D10]	Conceptual inputs; check vintages and concepts

6.3 Build new

None of the following exists elsewhere in the Budget Lab:

- the sector bridge and production accounts with named Q (§3.1)
- the legal-form split of capital income (§3.3)
- the CIT baseline bridge and incremental-rate logic (§3.3); Cost-Recovery supplies one input
- payout on a matched universe (§3.4)
- the ownership × wrapper matrix (§3.5)
- the asset ledger and the retirement balance/withdrawal module (§3.6)
- the ordered reconciliation (§3.7)
- the flow-reconciliation output and conservation checks (§3.9, §5)

These are also the stages the review identified as the research contribution, so building them is expected rather than a cost overrun.

6.4 Effect on the plan

Adopting Macro-Projections as the baseline is the one change with immediate schedule value: it moves most of weeks 1–2’s CBO data work to “check one column definition and pin a vintage.” It also implies a small interface change — declare Macro-Projections as a dependency alongside Tax-Data (this is the “revisit config/interfaces/ in Phase 1” item in v2_decisions.md Appendix A).

7. Out of scope for v2.0

Occupation exposure and displacement (D8); UI and spending responses; endogenous GDP, investment, and interest-rate feedback (the macro adapter is an input contract, not a model); behavioral responses to tax changes; multinational and firm-level corporate microsimulation; full lifecycle retirement and estate modeling; endogenous buybacks and portfolio choice. Each has a hook kept in v2.0 so it can be added (see review, “What belongs in Version 2 and what should wait”).

8. Questions for review

1. **Estimand.** Is “relative to CBO, 2030 focus, annual accounting” the right release contract [D6, D13]? Do we want any years past 2030 in the published tables?
 2. **Survey vintage [D1].** If the economists’ column (55/54/52) rather than the pooled column we used in v1.0 is the right one, do we issue an erratum for v1.0 or note it only in the v2.0 comparison?
 3. **Reference valuation rule [D10].** Zero-announcement revaluation as the reference case, capitalization as the sensitivity — or the reverse?
 4. **Ownership granularity [D9].** Aggregate shares by entity type only, or also by wealth group (which interacts with sensitivity 6)?
 5. **Distributional headline [D12].** Which panel leads — cash or economic income?
 6. **Reuse (§6).** (a) Adopt Macro-Projections as the v2 baseline module? (b) Should BLSMM’s AI scenarios supply v2’s GDP path (so one scenario definition drives both models), or only its debt/GDP step?
(c) Which incidence convention leads D12 — Tax-Simulator’s house rule (0→20% labor) or one of OTA / TPC / CBO?
 7. **Mixed income [D3].** Three conventions are in use (macro $\phi = 0.5$, Tax-Simulator’s 0.8 labor in distribution tables, Saez-Zucman in AI-Fiscal’s pass-through writer). Which one becomes the single ϕ ?
-

Appendix A — the \$100 design test

From the review, illustrative numbers only. An extra \$100 of C-corp economic profit pays \$20 of federal CIT. Of the remaining \$80, \$40 is paid as dividends and \$40 retained. Ownership: US taxable 50%,

traditional retirement 30%, foreign 20%. Household dividend rate 15%; no retirement withdrawals or realizations this year.

Destination	Dividends	Retained (attributed)	Total after entity tax
US taxable owners	\$20	\$20	\$40
Traditional retirement	\$12	\$12	\$24
Foreign owners	\$8	\$8	\$16
All owners	\$40	\$40	\$80

Federal receipts rise \$23 (\$20 CIT + \$3 dividend tax). US owners keep \$61 of after-tax economic income (\$24 of it accruing inside retirement accounts); foreign owners \$16. $\$23 + \$61 + \$16 = \100 . Taxable gains would need a valuation and realization rule — they cannot be read off the \$40 retained.